

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

**Duration : 1 Hour
Total Marks:50**

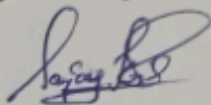
Annexure A

Analytical Problem Solving

Code of Conduct:

I Sarjay Krishnan R / 192521055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

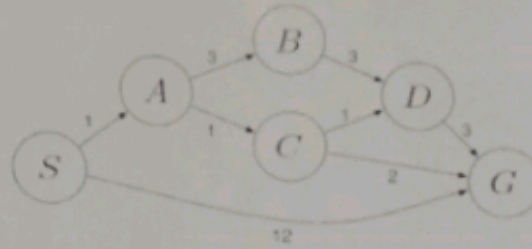


Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical Problem Solving1. Water Jug Problem:Problem Statement:

- Initial state: $(0,0)$ → Both jugs are empty
- First value → water in 4-gallon jug.
- Second value → water in 3-gallon jug.
- Goal state: $(2,0)$ → Exactly 2 gallons.

Operators:

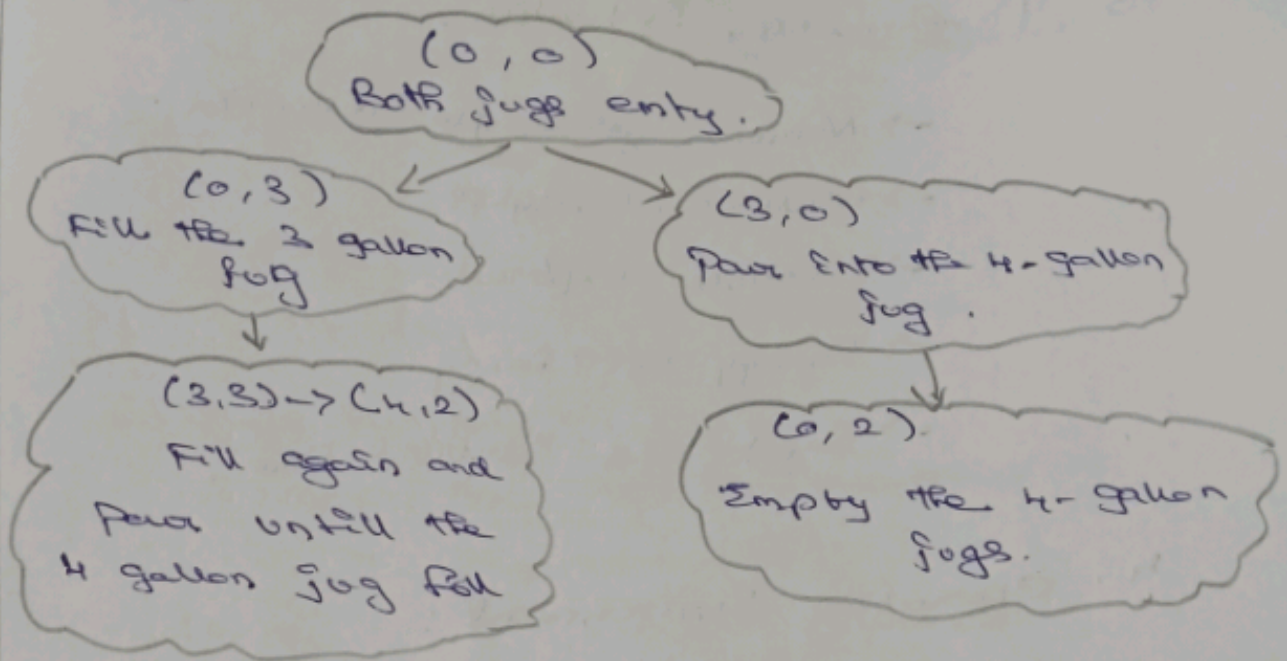
- * Fill a jug completely from pump.
- * Empty a jug on the ground.
- * pour water from one jug to another.

Problem description:

- Two jugs are available: one has a capacity of 4.
- water pump is available to fill the jugs.
- The water can be poured from one jug to another.

Steps	operations.	4 gallon jugs	3-gallon jugs.
1.	Initial both jugs are empty	0	0
2.	Fill the 3-gallon jug	0	3
3.	Pour water into the 4-gallon jug	3	0
4.	Fill the 3-gallon jug	3	3
5.	Pour into 4-gallon jugs.	4	3
6.	Empty the 4 gallon jug	0	2
7.	Pour the remaining water into 4 - gallon jug	2	0

State Transition Diagram.



2.

Mars Rover Agent.

1. Percept For agent:

- cameras (images of terrain and samples)
- sensors (temperature, pressure, atmosphere)
- chemical (sample composition)

2. Action of Mars:

- Move forward / backward
- Rotate left.
- collect soil and rock sample

3. Performance Evaluation.

- Number of sample collection.
- Accuracy analysis
- Distance explored
- Energy efficiency
- Safety and survival time.

4. Operating Environment:

- Calculation of time.
- Recovery of Systems.
- Energy Consumption.
- overviewing performance.

5. Agent Architecture:

- The rover has specific goal.

Eg: (Explore, Collect).

- It must select best action based on condition like ~~the~~ ~~energy~~.

Eg: Energy, Terrains, Scientific value

Queen Problem:

There are 5 Problem component in the Queen permutation.

1. Initial State:

It is a empty chessboard

2. State Representation:

Here the state is represented as

$[x_1, x_2, x_3, \dots, x_p]$

For eg:

$[1, 2, 8, 7, 2, 4, 6]$

3. Actions operations:

* Place the queen in the empty column

* Then move a queen to the another row.

4. Goal test:

→ No two queens are in the same row

→ No two queens are in the same column

→ No two queens share as a same diagonal.

5. Search Strategies:

* Depth First Search

* place queen column by a column.

4.

OLA cab booking mobile Application.

Problem used:

- > cost
- > Distance
- > Destination
- > Travel time
- > user preference.

Pseudocode:

START

Input Source location

Input destination location

Find available cabs

For each cab

Calculate distance

Calculate travel time

Select cab with max value

Customer select cab

confirm booking

Send delivery details

End.

Uniform Cost Search Algorithm (UCS)

Initial State:

Starting warehouses

Goal State:

Destination warehouse

Action:

Move from one to connected warehouse

Path cost:

It is the Total Level cost (Fuel + time).

USC:

It is a (Graph, Start, Goal)

Create priority queue.

Insert Start node with cost
while queue is not empty.

* Remove node

* IF node = Goal

Reserve path and cost.

Expand node.

Relation Failure.

Path : $S \rightarrow A \rightarrow C \rightarrow G$

Total cost = $\text{Cost}(S, A) + \text{Cost}(A, C) +$
 $\text{Cost}(C, G)$

